

ERICK REINALDO FLORES ZAMBRANO

Economist & Data Scientist | AI Engineer | ML Engineer

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PROFESSIONAL SUMMARY

Economist and Data Scientist specialized in AI and automation. Este es un resumen generado para comprobar que el texto dinámico se inyecta correctamente en el PDF ATS Premium.

KEY PROJECTS & PORTFOLIO

FCH-ARX V4 — Original Cryptographic Algorithm [Python, C, NIST FIPS 180-4]

- Designed and implemented a novel hash algorithm from scratch using ARX (Add-Rotate-XOR) operations with mathematically derived constants (Tesla 3-6-9, 26 rounds).
- Passed SAC-NIST test with 49.95% avalanche (SHA-256 reference: 50.02%). Speed: 185 MB/s in optimized C. Attack complexity: 2^{256} .

AI Job Hunter Bot V4 — Autonomous 24/7 Agent [Python, LinkedIn API, Gmail API, OpenRouter, GitHub Actions]

- Full automation pipeline: scraping 9+ job platforms, real email verification via Hunter.io API, personalized cover letters generated with LLMs.
- Automatic Gmail sending with CV attachment, anti-duplicate memory, and 24/7 deployment on GitHub Actions.

Algorithmic Trading with Deep Learning [PyTorch, LSTM, Backtrader, MLflow, Python]

- LSTM model for S&P 500 price forecasting achieving 68% directional accuracy.
- Complete pipeline: data ingestion, feature engineering, momentum/mean-reversion backtesting, experiment tracking with MLflow.

Multi-Agent System with MCP (Model Context Protocol) [Python, FastAPI, Docker, LLMs, GARCH]

- Distributed agent architecture for automated financial analysis: scraping agent (Alpha Vantage, TradingView), econometric agent (GARCH volatility models), LLM agent for executive report generation.

RESTful API for Predictive Models [FastAPI, Docker, Random Forest, GitHub Actions]

- ML inference service in production: credit risk classification (Random Forest), Docker containerization, automatic Swagger documentation, pytest testing, basic CI/CD with GitHub Actions.

TECHNICAL SKILLS

AI / ML:	Python (expert), scikit-learn, PyTorch, TensorFlow, LLMs, Prompt Engineering, Hugging Face
Data Engineering:	SQL (advanced), PostgreSQL, ETL Pipelines, Pentaho, Airflow (concepts), Spark (concepts)
MLOps / DevOps:	FastAPI, Docker, REST APIs, GitHub Actions CI/CD, MLflow, Git
Finance / Econometrics:	Time Series (ARIMA, GARCH), Panel Data, Causal Inference, VaR, Backtesting
Business Intelligence:	Power BI (DAX, Power Query), Excel VBA, Data Storytelling, SQL Server
Languages:	Spanish (Native), English (B2 — technical reading/writing proficient)

PROFESSIONAL EXPERIENCE

Commercial Advisor — Vamoret S.A. (Palmón Group)	Nov 2022 – Present
- Lead a team of 3 people in customer service and operational improvement. - Designed executive Power BI dashboards with real-time KPIs, RFM segmentation, and DAX forecasting — deployed in production. - Automated monthly inventory reporting with Excel VBA macros, reducing processing time by 60%. - Applied data analysis for pricing strategies and customer retention decisions.	
Commercial Advisor — Hularuss S.A. (PepsiCo)	Oct 2018 – Mar 2021
- Managed a portfolio of clients using data-driven strategies (Excel + Power BI). - Trained new advisors in negotiation techniques and NLP (Neuro-Linguistic Programming). - Implemented inventory rotation and merchandising optimization at points of sale.	

EDUCATION

B.Eng. Data Science & Artificial Intelligence — Universidad de Guayaquil	2023 – Present
B.A. Economics — Universidad Técnica de Manabí (UTM)	2021 – Present
★ Managing two simultaneous university degrees remotely — demonstrates exceptional time management, self-discipline, and commitment to continuous learning.	

CERTIFICATIONS

- Python for Finance & Algorithmic Trading — Udemy (40h)
- SQL for Big Data — IBM (40h)
- Deep Learning Specialization — Coursera (in progress)
- LLMs & Prompt Engineering — DeepLearning.AI (in progress)
- Digital Marketing & Web Analytics — Google Activate (50h)